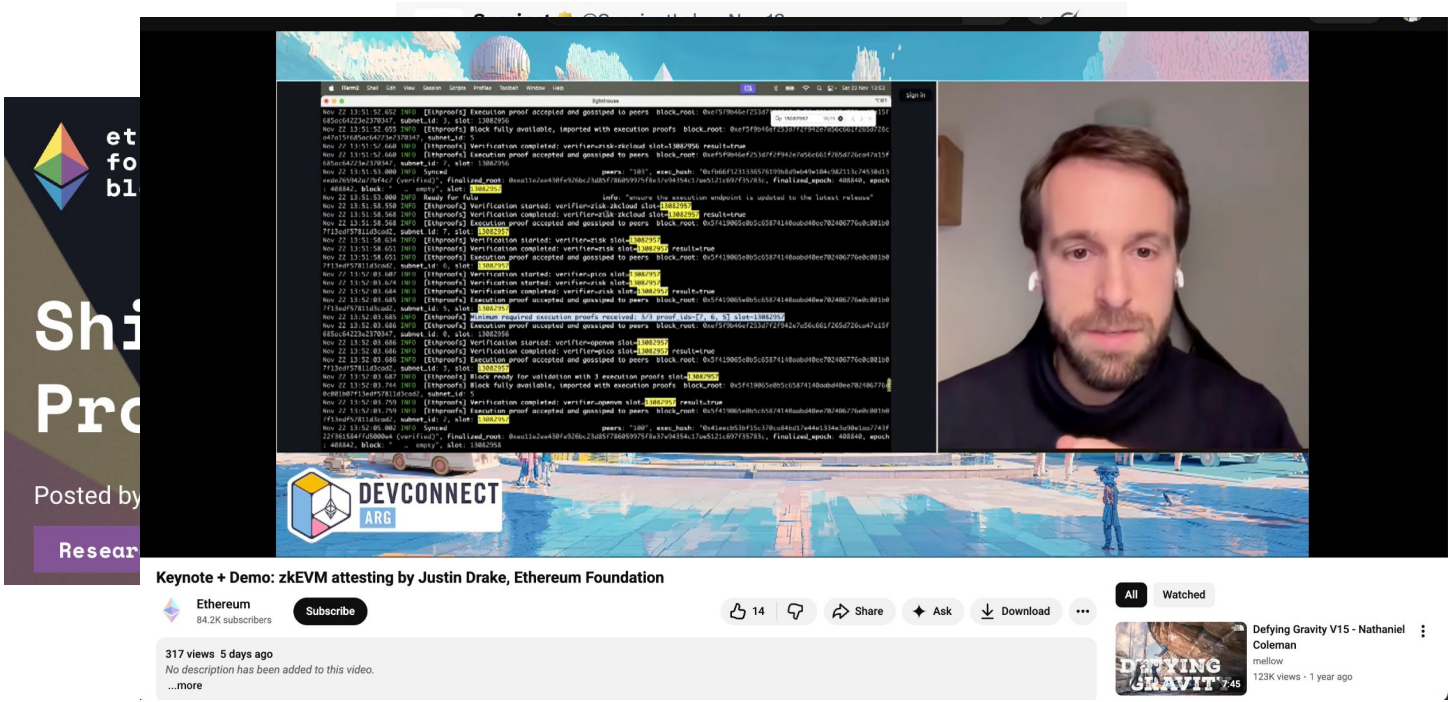
How to get untrusted people to work for you

mike neuder with Maryam Bahrani and Matt Weinberg
Princeton DeCenter Spring Conference 2026

Motivation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Research

Motivation

1. Each block will have a cryptographic proof of correctness.
2. The proof is expensive to make, but only one copy is needed.
3. Everyone else can just verify the proof.

Upshot

- Scale, scale, scale

Risks

- Protocol complexity
- Prover centralization
- Liveness attacks*

Research question:

“how can Ethereum permissionlessly acquire proofs?”

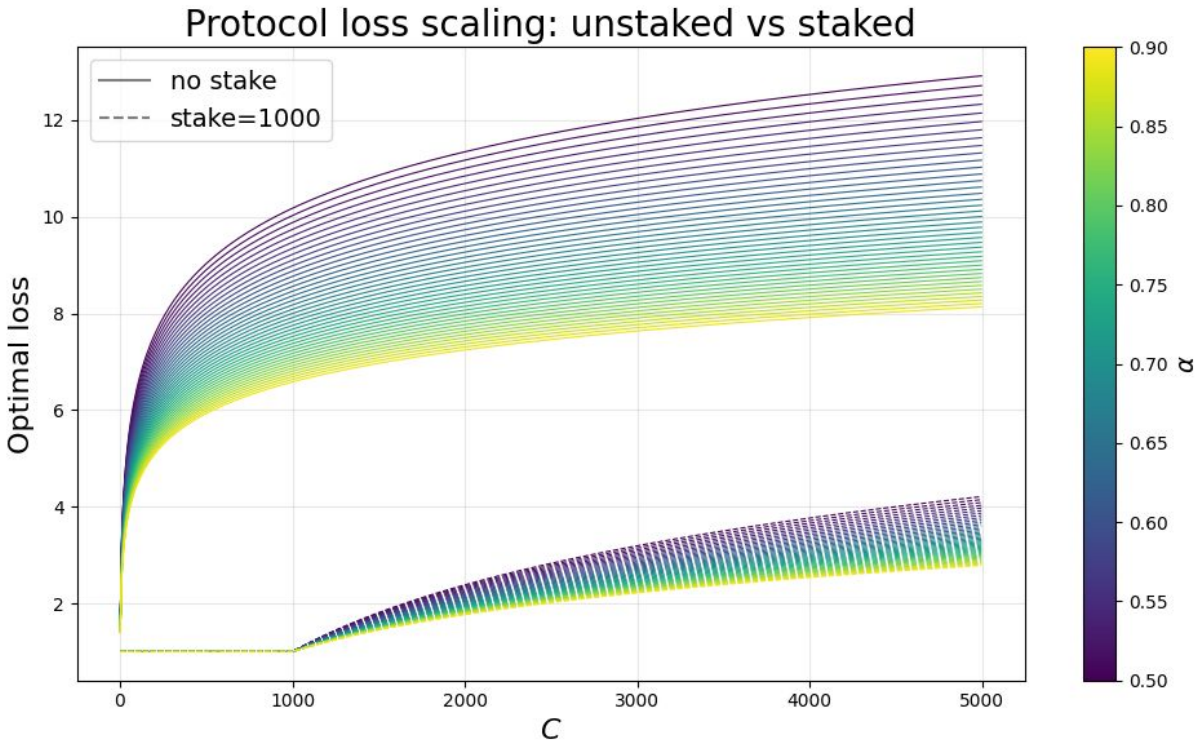
Model

- There are a set of n possible provers
- A Byzantine attacker can corrupt a subset of them
- The protocol only needs to get *a single copy* of the proof
- The proof is free to verify
- If no proof is delivered, then we have a “liveness fault”
- C = “The economic loss of a liveness fault”

Results

- We can get you asymptotically optimal cost of O(log C)
- The optimal mechanism has intuitive properties:
  - Choose one prover to be “main leader” for that round
  - Choose a committee of other provers to serve as “back up”
  - Condition everyone’s payments based on the leader
  - Staking can make the payments even lower O(1)!

Results


Protocol loss scaling: unstaked vs staked
no stake
stake=1000
Optimal loss
C
α
0
1000
2000
3000
4000
5000
2
4
6
8
10
12
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90

For practitioners

- C is big!
- Prover cost is small
- Stake is powerful

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For practitioners

- Use a rotating leader election as the primary mechanism
- Have a backup committee in case the leader no-shows
  - Use our mechanism to choose the committee size
- Require stake if possible

Thanks!